Fatgezim “Zim” Bela

Founder and CTO · Software Engineering · Data and Scientific Systems

fatgezimbela1@gmail.com | LinkedIn | GitHub | Portfolio

I lead technology and build software from workflow discovery through implementation and testing. My work spans typed web applications, Python and SQL data workflows, browser automation, and reproducible neuroscience software. A background in behavioral science and medical education helps me translate complex domain requirements into practical products.

Technical capabilities

Software and product building TypeScript and React, JavaScript and Next.js, Semantic HTML and CSS, Git and GitHub Actions, Vitest and Playwright, Accessible interface design, Testing and static deployment, Local-first data handling, Browser automation, Product workflow design, Responsible product prototyping

Data science and engineering Python, SQL, Statistical analysis, Regression, classification, and clustering, Data visualization, Workflow automation, Operational reporting

Project technologies Pynapple, NeMoS, Stan, PyTorch, Plotly, Zod, Vitest, Express, Firebase Auth, Firestore, PDFKit, docx

Experience

Founder and Chief Technology Officer | Create 13 Group

2018 - Present

Lead technology direction and hands-on software development, translating business requirements into product architecture and implementation priorities.

- Own technical scoping, integration decisions, interface development, and release planning.
- Work directly in the codebase while coordinating product requirements and iterative delivery.

Founder and Chief Technology Officer | NeuroPath LLC

2026 - Present

Lead product architecture and engineering for behavioral-health learning and operational software; co-build Bela Data Lab products with Meili Bela.

- Translate clinical and operational workflows into structured data, review states, and usable interfaces.
- Build browser-local analytics, education tools, and workflow automation with explicit data boundaries.

Data Engineer | The McVay Company, Inc.

October 2022 - January 2023

Contributed to data-engineering workflows at The McVay Company, Inc.

Board Certified Behavior Analyst

Since 2025

Behavior-analytic practice centered on assessment, individualized programming, skill development, caregiver collaboration, and team support.

- Functional behavior assessment and behavior intervention planning
- Skill-acquisition programming using evidence-based teaching approaches
- Caregiver collaboration and RBT supervision

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Selected engineering projects

NeuroStack Explorer Public source

Creator and Scientific Software Developer | Open-source scientific portfolio

An open-source neuroscience software explorer that turns a checksum-verified NWB dataset into readable, interactive views across data, models, and reproducible artifacts.

Stack Python, TypeScript, Pynapple, NeMoS, Stan, PyTorch, Plotly

Bela Behavior Data Lab Live application

Co-founder and Co-builder | Live operations product

A browser-local operations workspace that turns Rethink appointment exports into reviewable dashboards, follow-up views, and report-ready outputs.

Stack React, TypeScript, Vite, Clerk, Stripe, Supabase, SheetJS, jsPDF, browser-local data processing

RBT Practice Hub Live application

Founder and Product Builder | Live study resource

A static study application that organizes practice across the 19 RBT Initial Competency Assessment tasks with original review materials and local progress tracking.

Stack React, TypeScript, Vite, React Router, localStorage, GitHub Actions, Vercel

StepSpark Public source

Creator and Product Builder | Medical-learning prototype

A visual-learning prototype for rapid USMLE Step 1 recall, structured around source-aware cards, progressive disclosure, and learner-controlled review states.

Stack React, TypeScript, Vite, Zod, Storybook, Vitest, Playwright, accessibility testing, source and license metadata

Clinical Assessment Portal

Product Builder and Developer | Private implementation; locally tested with synthetic records

Implemented practice-scoped assessment workflows with ownership and edit permissions, version checks, clinician review and signatures, and server-generated PDF/DOCX reports.

Stack React, TypeScript, Express, Firebase Auth, Firestore, Zod, PDFKit, docx, Vitest

USMLE Sprint

Product Builder and Developer | Account-gated learning application; local implementation inspected

Built a study workspace with progress dashboards, daily planning, question-session controls, review history, and import/export workflows using JavaScript and browser persistence.

Stack HTML, CSS, JavaScript, localStorage, progress API integration

Education and credentials

Doctor of Medicine Candidate | Ross University School of Medicine | Expected 2028

M.Ed. in Special Education | George Mason University | 2020

Graduate Certificate in Applied Behavior Analysis | George Mason University | 2020

B.S. in Neuroscience | George Mason University | 2017

Board Certified Behavior Analyst (BCBA) | Licensed Behavior Analyst - Virginia | Licensed Behavior Analyst - North Carolina | Licensed Behavior Analyst - New York

Research

Computational Neuroscience: Tracing Neurons from the Fruit Fly Larva | Centers for Neural Informatics, Structures & Plasticity, George Mason University

Methods Fiji/ImageJ, Image-channel splitting, Signal combination, Vaa3D raw preparation, Vaa3D image stitching, NeuTube tracing, Vectorized-tree reconstruction